Discussion and Conclusion

How to develop an evidence-supported research argument and bring it to a clear close.

We begin with what the findings mean, examine explanations and their relationship to existing knowledge, then establish the contribution and synthesise the overall answer. Published paragraphs illustrate the methods after they have been explained.

Learning aims

By the end of today’s session, you will be able to:

1. Distinguish the purposes and content of the Discussion and Conclusion.
2. Develop interpretations and explanations appropriate to the findings and research design.
3. Connect findings to theory and prior research, including findings that differ from expectations or earlier studies.
4. Justify significance, contribution and implications while explaining the consequences of contradictory evidence and limitations.
5. Build a Conclusion that synthesises the overall answer and communicates a supported takeaway.

What Discussion and Conclusion each do

The same research question connects the sections. Results establish the evidence; the Discussion develops and examines its meaning; the Conclusion brings the overall answer into focus.

Section	Question it answers	How the content works
Results / Findings	What did the study find?	Present the patterns, estimates, themes or cases and the evidence needed to assess them.
Discussion	What do these findings mean, why might they have occurred, and what do they add?	Interpret particular findings, develop and test explanations, engage with theory and literature, and justify their significance and implications within the evidence's limits.
Conclusion	What is the study's overall answer and what should the reader retain?	Synthesise the central answer and contribution, preserve the scope that matters, and bring the takeaway into focus. Select the established ideas needed for that synthesis.

Some reports combine sections. Identify the intellectual work first, then use the structure required for the report. The learning route is the same: develop an answer → examine and position it → establish what it adds → bring the answer together.

How to plan a connected Discussion

Begin with the research questions and the important findings that answer them. Give each subsection a central interpretive claim, then develop the reasoning that makes it defensible.

How to develop it

1. **Group findings around the answer they help develop**
   Select the patterns that matter to each research question. Bring related findings together and give the central findings sufficient space.
2. **State what the finding helps us understand**
   Use that interpretation as the controlling idea of the paragraph or subsection. Recall the specific evidence needed to support it.
3. **Develop the reasoning around that idea**
   Explain possible reasons, use theory and literature to examine the interpretation, and consider contradictory evidence or limitations where they change the claim.
4. **Show why the answer matters**
   Identify what becomes clearer, the contribution to knowledge and the implications that follow. Connect the next subsection to the developing answer.

> **Check the reasoning:** Paragraph functions can be combined and reordered. A strong paragraph has a central idea and visible links between evidence, interpretation and explanation; a complex argument may need several paragraphs.

How to interpret a finding

Interpretation explains what a finding means for the research question. Choose the kind of meaning the evidence can support. Several routes may contribute to the same argument.

Route	How to develop the interpretation	What to check
Substantive meaning	Translate the observed pattern into an answer about the phenomenon. For numerical results, explain direction, magnitude and uncertainty in the context of the measure.	Does the interpretation retain the outcome, participants and precision of the finding?
Explanatory meaning	Identify a process that could account for the pattern. Explain the connection and state whether it is supported or still a possibility.	Which evidence bears on the process? Could other processes produce the same pattern?
Theoretical meaning	Relate the finding to a specific concept or proposition. Explain whether it supports, refines or challenges the expected relationship.	Is the concept appropriate, and was the relevant relationship actually examined?
Contextual or developmental meaning	Use contrasts across cases, settings or time to explain conditions, experiences or change.	What makes the cases or observations comparable? Where does the interpretation apply?
Synthetic meaning	Consider related findings together and state the understanding that their combination makes possible.	Can the findings be connected through linked cases, timing or a justified analytic argument?

> **Check the reasoning:** A p value, theme label or quotation needs an explanation of its substantive meaning. Distinguish the observed pattern, participants' accounts and the researcher's interpretation.

▼ **Original example: read and underline the authors' moves**

> **Study context:** a Chinese university writing course; one teacher and 33 students; automated writing evaluation (AWE), peer and teacher feedback; analysis of drafts, feedback and retrospective interviews.

Zhang & Hyland (2022) · accepted manuscript, PDF p. 23 · Publication details

This study reinforces the view that student engagement with feedback is a complex process (e.g., Handley et al., 2011; Yu et al., 2019; Zhang & Hyland, 2018) as it taps into three dynamically interconnected dimensions. With respect to behavioural and affective dimensions, not only do students need to devote time and invest effort in their writing and revision after receiving feedback, but they must also regulate emotional and attitudinal reactions toward feedback. Perhaps the most difficult part of engagement is its cognitive dimension as it requires students to employ cognitive and meta-cognitive strategies to evaluate feedback on their writing and monitor their revision. How to evaluate written texts, identify problems, and revise can pose tremendous challenges to students.

How the passage develops the argument

- **Interpretation:** The highlighted sentences state what the authors make of the findings. The rest of the paragraph develops that interpretation through relevant concepts or supporting evidence.
- **Theory:** Behavioural engagement concerns effort and action; affective engagement concerns feelings and attitudes; cognitive engagement concerns evaluating and monitoring. These concepts organise the interpretation.
- **Literature:** The authors identify earlier ideas relevant to the interpretation. The citations show which claims in this paragraph are connected to previous scholarship.
- **Problem:** The difficulty of evaluating and revising writing identifies a problem that an explanation of the feedback process needs to address.

How to develop and examine explanations

Once the meaning is clear, ask why the pattern might have occurred. Develop an explanation through a chain of reasoning and give plausible alternatives a fair examination.

How to develop it

1. **Formulate a possible process**
   Draw on theory, prior research, contextual knowledge and participants' accounts. Explain how the process could produce the finding, including the intermediate steps.
2. **Identify evidence that would distinguish explanations**
   Ask what each explanation would lead you to expect. Examine the relevant observations, case contrasts, estimates, time order or linked sources.
3. **Examine credible rivals and contrary evidence**
   Consider other influences, measurement explanations and cases that do not fit. Judge them using evidence appropriate to the design.
4. **State the evidential status**
   Explain which account is better supported, which remains plausible and which the study cannot distinguish. Identify a focused follow-up if decisive evidence is missing.

> **Check the reasoning:** An association establishes a relationship within the design's assumptions. A participant's reported reason is an account to interpret. A causal explanation needs a design and evidence that address alternative causes.

▼ **Original example: read and underline the authors' moves**

> **Study context:** a Chinese university writing course; one teacher and 33 students; automated writing evaluation (AWE), peer and teacher feedback; analysis of drafts, feedback and retrospective interviews.

Zhang & Hyland (2022) · accepted manuscript, PDF p. 23 · Publication details

We believe that the integrated approach in this study played a pivotal role in facilitating student engagement with feedback, writing, and revision. The teacher was aware of feedback research where AWE feedback is found to offer immediate diagnostic assessment and multiple drafting opportunities (Stevenson & Phakiti, 2014), and how peer feedback might reduce writing anxiety and promote collaborative learning (Yu et al., 2019). As a result, she integrated both to create a feedback-and-revision process that made student revision less daunting and more manageable. As illustrated by the students, the complementarity of AWE feedback and peer feedback enabled them to distribute their attention to grammar and content in different drafts.

How the passage develops the argument

- **Interpretation:** The highlighted sentences state what the authors make of the findings. The rest of the paragraph develops that interpretation through relevant concepts or supporting evidence.
- **Literature:** The authors identify earlier ideas relevant to the interpretation. The citations show which claims in this paragraph are connected to previous scholarship.
- **Explanation:** The authors connect the activities to manageable revision and attention to grammar and content. "We believe" signals their interpretation; the classroom case does not isolate a causal effect.
- **Evidence:** "As illustrated by the students" returns the explanation to students' reported revision experiences. The reader can see which finding supports the proposed process.

How to build a coherent account

An explanation becomes more useful when it connects the findings. Use theory or a conceptual account to clarify those connections, then show what the evidence does to that account. Inductive and descriptive studies can develop a grounded account without testing a named theory.

How to develop it

1. **Connect the relevant findings**
   Identify the shared question and explain how the findings complement, qualify or contradict each other. In mixed methods, show what the combination adds beyond the separate findings.
2. **Specify the conceptual relationship**
   Name the concept or proposition that explains the connection in plain language. State what pattern it would lead you to expect.
3. **Examine the relationship against the evidence**
   Explain whether the evidence supports the account, identifies a condition, challenges an assumption or suggests a new relationship to investigate.
4. **State the integrated understanding**
   Bring the findings into a substantive answer and retain differences that materially change it. Explain how the account reaches beyond the individual observations.

> **Check the reasoning:** A theory citation supplies a source; the argument must explain the link. Consistency with a theory supports a bounded interpretation, while stronger claims require stronger tests.

▼ **Original example: read and underline the authors' moves**

> **Study context:** a Chinese university writing course; one teacher and 33 students; automated writing evaluation (AWE), peer and teacher feedback; analysis of drafts, feedback and retrospective interviews.

Zhang & Hyland (2022) · accepted manuscript, PDF pp. 23–24 · Publication details

The integrated approach encouraged the students to focus on local and global areas at different stages of their writing and helped them realise that revision is an integral part of writing. Moreover, the purposeful and systematic combination of AWE, peer, and teacher feedback helped enriched students' learning experiences by offering varied learning modes (human and machine) and diverse readership (peer and teacher). It is important to emphasise the role of teachers in the feedback cycle. As this study shows, the teacher's active involvement in both AWE and peer feedback, by asking students to focus on improving their scores at the AWE stage and produce a reflective report after peer feedback contributed to students' active engagement through a sense of progress and reflection. Without the final step of teacher feedback, however, the integrated approach would probably have been less effective as many of the students commented on its importance in their writing.

How the passage develops the argument

- **Synthesis:** These sentences bring different observations into an account of the writing-and-revision process. The paragraph gives their relationship a meaning.
- **Central interpretation:** The teacher's role gives the paragraph a central interpretive point linking the activities together.
- **Evidence for the connection:** The authors refer to the teacher's actions and students' comments as evidence for how support operated.
- **How the process may work:** Varied modes, a sense of progress and reflection are used to explain the observed engagement. These mechanisms remain the authors' interpretation.
- **Condition on the explanation:** The final sentence considers what might happen without teacher feedback. It identifies a proposed condition; the study did not experimentally remove that step.

How to engage with existing literature

The developing answer now enters the scholarly conversation. Choose the specific earlier claim relevant to your finding, establish what is comparable, and explain how the relationship changes the interpretation.

Relationship	How to develop the comparison	What the reader should learn
Supports	Identify the shared pattern and show that the constructs, outcomes and conditions are sufficiently comparable.	How the present evidence strengthens confidence in a particular understanding.
Extends	State what earlier work established, then identify the additional process, setting, period or outcome examined here.	What becomes clearer beyond the earlier account and why that addition matters.
Qualifies	Identify the condition or distinction that changes when or for whom the earlier claim holds.	A more precise account of the relationship and its boundaries.
Challenges	Establish the actual discrepancy, examine its robustness and compare plausible methodological or substantive explanations.	Which claim needs reconsideration and what remains unresolved.

How to develop it

1. **Build the comparison around an idea**
   State the earlier claim, relate the current finding to it, and explain the meaning of that relationship. Use citations to support each step.
2. **Check the basis for comparison**
   Inspect the relevant methods and results. Distinguish differences in concepts, measures, populations, tasks, settings and timing from differences in the substantive phenomenon.

> **Check the reasoning:** The explanation of the relationship carries the argument. Agreement alone needs a reason to matter; disagreement needs investigation before a stronger conclusion follows.

When findings differ from earlier research

A difference is a question to investigate. Work out what differs, examine why, and let the available evidence determine how far the explanation can go.

How to develop it

1. **Establish the discrepancy precisely**
   Identify the difference in direction, magnitude, experience, process or condition. Check estimates and uncertainty. A significant result in one study and a nonsignificant result in another do not by themselves establish a difference between the studies.
2. **Check whether the studies address the same question**
   Compare the construct, outcome, measure, participants, task, setting, design and time period. Different questions can produce different findings without directly contradicting each other.
3. **Develop and weigh possible explanations**
   Separate methodological possibilities from substantive conditions. Use evidence from the studies to explain how each difference could produce the observed contrast.
4. **Reach a proportionate interpretation**
   If a contextual explanation is supported, specify the condition it reveals. If the studies cannot distinguish explanations, state the unresolved discrepancy and the evidence needed to resolve it.

> **Check the reasoning:** Label an untested explanation as a possibility. A new setting becomes a contribution when the comparison shows what that setting changes in the understanding.

▼ **Original example: read and underline the authors' moves**

> **Study context:** interviews with ten higher-education teachers about the knowledge, skills and dispositions they use to make technology-enhanced feedback more relational. Their curriculum responsibilities varied.

Payne, Ajjawi, & Holloway (2023) · accepted manuscript, PDF p. 19 · Publication details

Carless and Winstone's (2020) design dimension is differently positioned; their dimension focuses on the ways in which teachers contribute to effective feedback processes through their design of curricula and assessment. This positioning appears to be aligned with Boud and Dawson's (2021) macro-level development and/or meso-level coordination and staging responsibilities. Many of our participants suggested that the feedback process would be improved if it could be made more dialogic and formative, indicating that sequenced assessments may better enable students to use feedback from one assignment to the next. However, due to our participants' varied levels of responsibility they were not all responsible for curriculum design. This may explain the difference in strategies that we identified compared with Carless and Winstone (2020). Instead, our design literacy strategies were aimed at managing the tensions between grading and feedback (Boud and Dawson 2021).

How the passage develops the argument

- **Earlier framework:** The authors identify the specific design dimension they are engaging with and locate it at curriculum and assessment level.
- **Current findings:** The participants' accounts introduce what this study found about the possibilities for dialogic, formative feedback.
- **Relevant context:** Different levels of curriculum responsibility provide a concrete contextual distinction. It gives the comparison an evidential basis.
- **Explanation of the difference:** "This may explain" explicitly connects the contextual distinction to the difference between studies. "May" retains the explanation's tentative status.
- **What the comparison adds:** The final sentence states the focus of the design strategies identified here. The comparison makes that focus more precise.

How to handle exceptions and limitations

The same scrutiny applies within the study. Contrary evidence tests the explanation; a limitation affects a particular inference. Work through their consequences so the reader can judge the answer's reach.

What needs attention	How to reason through it	How the claim may change
A case or finding that does not fit	Check its credibility and context, compare it with the broader evidence, and examine whether it reveals a condition or another process.	Refine the explanation, restrict its reach or reconsider it if the contrary evidence undermines the central claim.
A design, data or analysis limitation	Name the affected inference and explain how the constraint changes confidence in the claim. Specify what the evidence still supports.	Qualify the causal, measurement, process or transfer claim that is affected; preserve the supported understanding.

How to develop it

1. **Make the consequence explicit**
   Connect the evidence or constraint to the specific claim. Explain why it matters to the main argument and how consequential it is.
2. **Link any follow-up to the uncertainty**
   Propose the observation, comparison, measure or design that would address the remaining question. Keep the proposal proportionate to what is unresolved.

> **Check the reasoning:** A list of limitations leaves their effect unclear. Explain the affected inference and its consequence at the relevant point in the argument.

▼ **Original example: read and underline the authors' moves**

> **Study context:** a Chinese university writing course; one teacher and 33 students; automated writing evaluation (AWE), peer and teacher feedback; analysis of drafts, feedback and retrospective interviews.

Zhang & Hyland (2022) · accepted manuscript, PDF pp. 24–25 · Publication details

We believe that the integrated approach is not a panacea to poor engagement of some students and that individual factors, such as motivation and language proficiency, can also colour the picture (Zhang & Hyland, 2018). Indeed, there were students in this study that failed to engage and responded with indifference to the multiple feedback types, seeing the revision cycle as overkill and inefficacious. Not all students will be sufficiently incentivised to participate in what might seem a trick to get them to work harder on the same piece of writing. This is one challenge that lies in enacting this approach. However, we believe that enough students were encouraged by the teacher's obvious enthusiasm for the approach, her interest in her students' improvement, and her use of supportive teaching methods to make improvement. Together this meant that the approach, in this situation at least, helped to promote student involvement and address learners' different needs.

How the passage develops the argument

- **Qualification:** The authors make their interpretation conditional and acknowledge influences beyond the feedback approach.
- **Contrasting evidence:** Students who failed to engage provide evidence that the approach did not operate in the same way for everyone.
- **Alternative influences:** The paragraph considers motivation, the perceived demands of revision and teacher support. These are possible influences to examine, not isolated causal effects.
- **Scope of the claim:** "In this situation at least" explicitly restricts the synthesis to the studied setting.
- **Practical consequence:** The challenge in enacting the approach follows directly from the contrasting evidence.

▼ **Original example: read and underline the authors' moves**

> **Study context:** interviews with ten higher-education teachers about the knowledge, skills and dispositions they use to make technology-enhanced feedback more relational. Their curriculum responsibilities varied.

Payne, Ajjawi, & Holloway (2023) · accepted manuscript, PDF p. 21 · Publication details

Our research question centred on teacher feedback literacies and strategies required to implement relational technology-enhanced feedback. We interviewed ten higher education teachers across a range of disciplines and cohorts with technology-enhanced feedback experience to understand their reported feedback literacies. One limitation of this approach is that there might be discrepancies between participants' actual feedback practices and what they reported. It was not our intention to measure nor make claims of the efficacy of technology-enhanced feedback nor its ability to improve academic performance. It was also not our intention to produce generalisable findings, thus there may be additional literacies and strategies not identified here. Qualitative research samples are small, for good reason, and are justified when good purposive sampling has taken place (Ritchie et al. 2014).

How the passage develops the argument

- **Research question:** The authors restate the question so readers can judge the intended reach of the study.
- **Evidence collected:** Interviews establish that the study concerns teachers' reported literacies.
- **Limitation and its consequence:** Reported practices may differ from actual practices. The limitation concerns what interviews reveal about action.
- **Claim boundaries:** The authors identify claims about efficacy, academic performance and generalisability that their study was not intended to make.
- **Design rationale:** The final sentence explains the rationale for purposive qualitative sampling. It preserves the design's purpose while acknowledging its scope.

How to develop significance and contribution

Once the answer has been examined, establish what it adds. Contribution identifies the particular addition to knowledge; significance explains why that addition matters for a problem, concept or decision.

How to develop it

1. **Return to the established problem**
   Use the Introduction and relevant literature to state what was understood before and what remained uncertain.
2. **Identify the addition the findings support**
   Specify the new pattern, more credible estimate, process, distinction, condition, theoretical refinement or methodological insight. Connect it to the actual evidence.
3. **Explain the change in understanding**
   Show how the addition makes the previous account more complete, precise or appropriately bounded. This is how the contribution advances the argument.
4. **Explain why the change matters**
   Identify the scholarly question or practical decision affected and explain the consequence. Keep the importance proportional to the design and evidence.

> **Check the reasoning:** A new location or a claim to “fill a gap” needs an explanation of what becomes clearer. Substantive significance concerns importance; statistical significance concerns a statistical test.

▼ **Original example: read and underline the authors' moves**

> **Study context:** a Chinese university writing course; one teacher and 33 students; automated writing evaluation (AWE), peer and teacher feedback; analysis of drafts, feedback and retrospective interviews.

Zhang & Hyland (2022) · accepted manuscript, PDF p. 24 · Publication details

By following a teacher that integrates three types of feedback we have sought to move away from simplistic dichotomies about the 'best' type and learnt that there is no single best approach to written feedback. The integrated approach responds to those who question the legitimacy of computer-generated diagnostics (e.g., Stevenson & Phakiti, 2014; Ware, 2011) by supporting it with peer and teacher feedback. It creates positive synergistic effect that increases student involvement and motivation. Overall, the integration of three types of feedback seems to offer the following benefits:

1. encourage the students to devote time and effort in their writing and revising, which is an effective way to improve their behavioural engagement.
2. reduce the demotivating wait for timely feedback and offers peer support for writing development, promoting affective engagement.
3. require students to analyse, evaluate, monitor, and regulate their writing and revisions, thus drawing on cognitive strategies.

How the passage develops the argument

- **Contribution:** The opening identifies a change in understanding: considering how feedback types are combined. A contribution claim needs this specific addition, supported by the study.
- **Scholarly issue:** The next sentence positions that addition in relation to concerns about computer-generated feedback.
- **Why it matters:** The claimed benefits explain why the combined approach matters for involvement, motivation and different dimensions of engagement. Here significance concerns importance, not a statistical test.
- **How the importance is developed:** The paragraph introduces a set of connected benefits and links each to the engagement concepts. The authors develop the importance through those relationships.

How to derive justified implications

Implications develop what follows from the answer. Trace each recommendation to a finding and explain why the proposed response is warranted.

How to develop it

1. **Identify the finding and its consequence**
   State the problem or possibility made visible by the evidence and explain the reasoning connecting it to a response.
2. **Name the actor and action**
   For practice, specify who could do what. For research, specify the unresolved question and the study that could address it.
3. **State the conditions**
   Explain the setting, resources, support or assumptions needed for the response to make sense. Consider who or what falls outside its scope.
4. **Propose an appropriate level of action and evaluation**
   An untested suggestion can be piloted and assessed. Stronger recommendations require evidence of effectiveness and applicability. Explain what evidence would show whether the proposed action helps.

> **Check the reasoning:** The action should address the problem identified by the findings. “More research is needed” becomes useful when it names the uncertainty and a way to investigate it.

▼ **Original example: read and underline the authors' moves**

> **Study context:** a Chinese university writing course; one teacher and 33 students; automated writing evaluation (AWE), peer and teacher feedback; analysis of drafts, feedback and retrospective interviews.

Zhang & Hyland (2022) · accepted manuscript, PDF p. 25 · Publication details

For teachers, promoting student engagement with feedback requires skills and awareness, involving the creation of inclusive and cooperative learning conditions. As shown in this study, the heavy assessment duties and marking load, described by Harland and Wald (2020) as 'assessment arms race', are not uncommon in many higher education contexts across the globe (Boud & Molloy, 2013). The fact that the teacher was able to integrate different types of feedback in her class without major alterations to classroom arrangements shows that her approach is worth exploring in other classrooms where writing, revision, and assessment are important components. Moreover, with growing time constraints, student numbers and pressures on academics to engage in research, administration and outreach, this integrated approach may offer a much-needed solution to student dissatisfaction with their feedback experiences in many parts of the world.

How the passage develops the argument

- **Conditions for action:** The opening identifies the skills, awareness and learning conditions needed to support engagement.
- **Problem the action addresses:** Assessment duties and marking load establish a practical problem for the proposed response.
- **Connection to wider research:** The literature places the classroom problem within a broader discussion of assessment workload.
- **Reason for the recommendation:** The observed ability to integrate feedback without major classroom changes provides the argument for exploring the approach elsewhere.
- **Proposed implication:** The authors propose exploration in other classrooms and a broader possible benefit. These extensions need local adaptation and evaluation; they are not established effects in those settings.
- **Practical importance:** The final sentence explains why the suggestion might matter under wider pressures on academics.

How to construct the Conclusion

The Discussion has developed the answer in detail. The Conclusion now brings the established reasoning together at the level of the whole study. Select the ideas needed to communicate what is now understood and why it matters.

Function	How to develop it	How it differs from the Discussion
Overall answer	Return to the central research question and combine the main findings into a direct, substantive answer.	Select the understanding reached after the detailed interpretations have been examined.
Contribution and importance	State the particular addition to knowledge and why that addition matters.	Foreground the central contribution already justified in the Discussion.
Necessary scope	Retain the condition or uncertainty that materially affects the overall answer.	Carry forward the key qualification while keeping the detailed evaluation in the Discussion.
Warranted consequence	Briefly state the key implication or focused next question if it is needed to understand the answer's significance.	Use consequences that have already been developed and justified.
Embedded takeaway	Connect these ideas into the central understanding the reader should retain.	Bring the argument to a clear close through synthesis.

> **Check the reasoning:** Adapt these functions to the study's needs. Read the Conclusion against the Introduction and final Discussion claims: the answer, contribution and scope should align.

How the Conclusion delivers its takeaway

A takeaway is the central understanding the reader should retain after the details. A summary selects the main points; synthesis explains what they amount to together. The takeaway is embedded in that synthesis and can develop across the ending paragraph.

How to develop it

1. **Choose the central understanding**
   Identify the answer that connects the important findings to the original problem.
2. **Keep the relationship that makes it informative**
   State the process, distinction or condition that the study has clarified and why it matters. This gives the ending substantive content.
3. **Retain the scope that keeps it accurate**
   Keep a necessary qualification within the synthesis. Make every part traceable to the evidence and reasoning already developed.
4. **Let the ending crystallise the answer**
   Read the Conclusion as a connected whole. Its final statement should make the established understanding clear to a reader who will not retain every detail.

> **Check the reasoning:** If the ending needs a new result, causal claim or undeveloped recommendation to work, revise the claim or develop the missing argument earlier. In the example, read all three paragraphs together before selecting “Takeaway”.

▼ **Original example: read and underline the authors' moves**

> **Study context:** a Chinese university writing course; one teacher and 33 students; automated writing evaluation (AWE), peer and teacher feedback; analysis of drafts, feedback and retrospective interviews.

Zhang & Hyland (2022) · accepted manuscript, PDF pp. 25–26 · Publication details

Overall, our study shows that student engagement with feedback can be facilitated with an integrated approach and the integration of different types of feedback merits attention from both researchers and practitioners in higher education. We have sought to bring the fields of assessment in higher education and L2 writing together in our study. These are two fields where feedback takes a prominent role. We have done this by tracking changes in student drafts, a method commonly used in L2 writing research but less common in assessment in higher education. Together with self-report data, we hope this has helped to shed some light on student engagement with feedback. The challenges of extrapolating to other classrooms and populations should be recognised and our conclusions seen as suggestive.

We recognise, of course, that a case study has limited generalisability and reflects the thoughts and practices of individuals in one context. It does, however, allow us to explore these thoughts and practices in greater detail to raise important issues which may escape large scale surveys while illuminating points which may be relevant for teachers in other situations. We hope this encourages others to reflect on the possibilities our study suggests and explore how the use of different types of feedback might promote greater engagement among their own students, adapting the approach to their own classroom context and constraints. We also acknowledge that there is space for future studies to adopt other methodological approaches to measure cognitive engagement through, for example, think-aloud protocols and eye-tracking software.

Most importantly, we argue that student engagement with feedback is a complex issue and that we cannot change students overnight. We can, however, develop inclusive pedagogies and create supportive learning environments that facilitate their engagement in writing and revision. By integrating multiple types of feedback – AWE, peer and teacher feedback – teachers might not only offer students a more varied writing experience that accommodates different learning styles and needs, but also facilitate active redrafting and improved writing through greater behavioural, affective and cognitive engagement. This, then, is an effective means of using assessment to feedforward: turning feedback into learning.

How the passage develops the argument

- **Overall answer:** The opening and synthesis state the answer reached through the study. They select the central understanding without repeating each individual finding.
- **Contribution:** The authors identify the connection between research fields and the use of draft changes alongside self-report. This explains the particular addition made by the study.
- **Importance:** The opening identifies why the answer merits researchers' and practitioners' attention.
- **Scope and uncertainty:** The authors retain the classroom setting, limited generalisability and the difficulty of extrapolation. These qualifications make the overall answer accurate.
- **What follows:** The authors invite adaptation and exploration in other settings and describe supportive pedagogical possibilities. These implications have grown out of the preceding argument.
- **Future research:** The suggestion of other methods addresses the remaining difficulty of examining cognitive engagement. A useful next study follows from an identified uncertainty.
- **Takeaway within the Conclusion:** The final paragraph develops the understanding to retain: engagement is complex, and supportive feedback processes can help turn feedback into learning. The last sentence crystallises that synthesis; its force comes from the paragraphs before it.

Thank you